Tejas Gokhale

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Assistant Professor; University of Maryland, Baltimore County

Current Position

University of Maryland, Baltimore County

– Assistant Professor, Department of Computer Science and Electrical Engineering 2023–
– Affiliate Faculty, Center for Artificial Intelligence 2024–
– Affiliate Faculty, Public Health Research Center 2026–
– Director, Cognitive Vision Group 2023–

Education

Arizona State University 2018–2023
Doctor of Philosophy (Ph.D.), Computer Engineering
Advisors: Yezhou Yang, Chitta Baral
Thesis: Towards Reliable Semantic Vision

Carnegie Mellon University 2016–2017
Master of Science, Electrical and Computer Engineering
Advisor: Aswin Sankaranarayanan

Birla Institute of Technology and Science, Pilani 2011–2015
Bachelor of Engineering (Honours), Electronics and Instrumentation Engineering

Previous Affiliations

Johns Hopkins University Summer 2026
Visiting Scholar, Human Language Technology Center of Excellence

Microsoft Research Summer 2022
Research Intern, Adaptive Systems and Interaction Group
Mentors: Hamid Palangi, Besa Nushi, Vibhav Vineet, Eric Horvitz

Lawrence Livermore National Laboratory Summer 2021, 2020
Research Scholar, Machine Intelligence Group
Mentors: Rushil Anirudh, Jay Thiagarajan, Bhavya Kailkhura

Arizona State University
Graduate Research Associate, School of Computing and AI 2018–2023
Graduate Teaching Associate, School of Computing and AI 2018–2020

Snap Research Summer 2018
Research Intern, Computational Imaging Group
Mentors: Guru Krishnan, Shree Nayar

Carnegie Mellon University 2017–2018
Graduate Student Researcher, Dept. of Electrical and Computer Engineering

ST Microelectronics Fall 2014
Intern, High Speed Links Group

Steel Authority of India Limited Summer 2013
Summer Intern, Bhilai Steel Plant

Publications

My Google Scholar page has updates and citation information. Citations: 1200+, h-index: 19, i10-index: 23

Peer-Reviewed Conference Proceedings

[C27] Shivanand Kundargi, Kowshik Thopalli, and Tejas Gokhale. "Sequentially Acquiring Concept Knowledge to Guide Continual Learning". In: *Conference on Lifelong Learning Agents*. PMLR. 2026
to-appear CoLLAs 2026

[C26] Sourajit Saha and Tejas Gokhale. "Zero-Shot Multimodal Retrieval with Multi-Scale Contextual Representations". In: *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2026, pp. 20304–20324
https://aclanthology.org/2026.acl-long.930/ ACL 2026

[C25] Naresh Kumar Devulapally, Shruti Agarwal, Tejas Gokhale, and Vishnu Suresh Lokhande. "Latent Diffusion Unlearning: Protecting Against Unauthorized Personalization Through Trajectory Shifted Perturbations". In: *Proceedings of the 33rd ACM International Conference on Multimedia*. 2025, pp. 11376–11384
https://arxiv.org/abs/2510.03089 ACM Multimedia 2025

[C24] Shaswati Saha, Sourajit Saha, Manas Gaur, and Tejas Gokhale. "Side Effects of Erasing Concepts from Diffusion Models". In: *Findings of the Association for Computational Linguistics: EMNLP 2025*. 2025, pp. 14991–15007
https://arxiv.org/abs/2508.15124 EMNLP Findings 2025

[C23] Nilay Yilmaz, Maitreya Patel, Yiran Lawrence Luo, Tejas Gokhale, Chitta Baral, Suren Jayasuriya, and Yezhou Yang. "VOILA: Evaluation of MLLMs For Perceptual Understanding and Analogical Reasoning". In: *The Thirteenth International Conference on Learning Representations*. 2025
https://arxiv.org/abs/2503.00043 ICLR 2025

[C22] Sourajit Saha and Tejas Gokhale. "Improving shift invariance in convolutional neural networks with translation invariant polyphase sampling". In: *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE. 2025, pp. 620–629
https://arxiv.org/abs/2404.07410 WACV 2025

[C21] Maitreya Patel, Naga Sai Abhiram Kusumba, Sheng Cheng, Changhoon Kim, Tejas Gokhale, Chitta Baral, et al. "Tripletclip: Improving compositional reasoning of clip via synthetic vision-language negatives". In: *Advances in neural information processing systems*. Vol. 37. 2024, pp. 32731–32760
https://arxiv.org/abs/2411.02545 NeurIPS 2024

[C20] Agneet Chatterjee, Gabriela Ben Melech Stan, Estelle Aflalo, Sayak Paul, Dhruba Ghosh, Tejas Gokhale, Ludwig Schmidt, Hannaneh Hajishirzi, Vasudev Lal, Chitta Baral, et al. "Getting it right: Improving spatial consistency in text-to-image models". In: *European Conference on Computer Vision*. Springer Nature Switzerland Cham. 2024, pp. 204–222
https://arxiv.org/abs/2404.01197 ECCV 2024

[C19] Agneet Chatterjee, Yiran Luo, Tejas Gokhale, Yezhou Yang, and Chitta Baral. "Revision: Rendering tools enable spatial fidelity in vision-language models". In: *European Conference on Computer Vision*. Springer Nature Switzerland Cham. 2024, pp. 339–357
https://arxiv.org/abs/2408.02231 ECCV 2024

[C18] Agneet Chatterjee, Tejas Gokhale, Chitta Baral, and Yezhou Yang. "On the robustness of language guidance for low-level vision tasks: Findings from depth estimation". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 2794–2803
https://arxiv.org/abs/2404.08540 CVPR 2024

[C17] Maitreya Patel, Tejas Gokhale, Chitta Baral, and Yezhou Yang. "Conceptbed: Evaluating concept learning abilities of text-to-image diffusion models". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. 13. 2024, pp. 14554–14562
https://arxiv.org/abs/2306.04695 AAAI 2024

[C16] Sheng Cheng, Tejas Gokhale, and Yezhou Yang. "Adversarial bayesian augmentation for single-source domain generalization". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023, pp. 11400–11410
https://arxiv.org/abs/2307.09520 ICCV 2023

[C15] Man Luo, Zhiyuan Fang, Tejas Gokhale, Yezhou Yang, and Chitta Baral. "End-to-end knowledge retrieval with multi-modal queries". In: *61st Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics (ACL). 2023, pp. 8573–8589
https://arxiv.org/abs/2306.00424 ACL 2023

[C14] Tejas Gokhale, Rushil Anirudh, Jayaraman J Thiagarajan, Bhavya Kailkhura, Chitta Baral, and Yezhou Yang. "Improving diversity with adversarially learned transformations for domain generalization". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2023, pp. 434–443
https://arxiv.org/abs/2206.07736 WACV 2023

[C13] Maitreya Patel, Tejas Gokhale, Chitta Baral, and Yezhou Yang. "CRIPP-VQA: Counterfactual reasoning about implicit physical properties via video question answering". In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. 2022, pp. 9856–9870
https://arxiv.org/abs/2211.03779 EMNLP 2022

[C12] Tejas Gokhale, Abhishek Chaudhary, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. "Semantically Distributed Robust Optimization for Vision-and-Language Inference". In: Association for Computational Linguistics, 2021, pp. 1493–1513
https://arxiv.org/abs/2110.07165 ACL Findings 2022

[C11] Tejas Gokhale, Swaroop Mishra, Man Luo, Bhavdeep Singh Sachdeva, and Chitta Baral. "Generalized but not Robust? Comparing the Effects of Data Modification Methods on Out-of-Domain Generalization and Adversarial Robustness". In: *Findings of the Association for Computational Linguistics: ACL 2022*. Association for Computational Linguistics, 2022, pp. 2705–2718
https://arxiv.org/abs/2203.07653 ACL Findings 2022

[C10] Neeraj Varshney, Pratyay Banerjee, Tejas Gokhale, and Chitta Baral. "Unsupervised Natural Language Inference Using PHL Triplet Generation". In: Association for Computational Linguistics, 2021, pp. 2003–2016
https://arxiv.org/abs/2110.08438 ACL Findings 2022

[C9] Yiran Luo, Pratyay Banerjee, Tejas Gokhale, Yezhou Yang, and Chitta Baral. "To Find Waldo You Need Contextual Cues: Debiasing Who's Waldo". In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Association for Computational Linguistics. 2022, pp. 355–361
https://arxiv.org/abs/2203.16682 ACL 2022

[C8] Man Luo, Arindam Mitra, Tejas Gokhale, and Chitta Baral. "Improving biomedical information retrieval with neural retrievers". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2022, pp. 11038–11046
https://arxiv.org/abs/2201.07745 AAAI 2022

[C7] Pratyay Banerjee, Tejas Gokhale, Yezhou Yang, and Chitta Baral. "Weakly supervised relative spatial reasoning for visual question answering". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2021, pp. 1908–1918
https://arxiv.org/abs/2109.01934 ICCV 2021

[C6] Pratyay Banerjee, Tejas Gokhale, Yezhou Yang, and Chitta Baral. "WeaQA: Weak Supervision via Captions for Visual Question Answering". In: *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021* (2021), pp. 3420–3435
https://arxiv.org/abs/2012.02356 ACL Findings 2021

[C5] Pratyay Banerjee, Tejas Gokhale, and Chitta Baral. "Self-Supervised Test-Time Learning for Reading Comprehension". In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics. 2021, pp. 1200–1211
https://arxiv.org/abs/2103.11263 NAACL 2021

[C4] Tejas Gokhale, Rushil Anirudh, Bhavya Kailkhura, Jayaraman J Thiagarajan, Chitta Baral, and Yezhou Yang. "Attribute-Guided Adversarial Training for Robustness to Natural Perturbations". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 35. 9. 2021, pp. 7574–7582
https://arxiv.org/abs/2012.01806 AAAI 2021

[C3] Tejas Gokhale, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. "Mutant: A training paradigm for out-of-distribution generalization in visual question answering". In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. 2020, pp. 878–892
https://arxiv.org/abs/2009.08566 EMNLP 2020

[C2] Zhiyuan Fang, Tejas Gokhale, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. "Video2commonsense: Generating commonsense descriptions to enrich video captioning". In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. 2020, pp. 840–860
https://arxiv.org/abs/2003.05162 EMNLP 2020

[C1] Tejas Gokhale, Pratyay Banerjee, Chitta Baral, and Yezhou Yang. "VQA-LOL: Visual question answering under the lens of logic". In: *European conference on computer vision*. Springer. 2020, pp. 379–396
https://arxiv.org/abs/2002.08325 ECCV 2020

Peer-Reviewed Journals and Magazines

[J1] Tejas Gokhale. "Towards robust visual understanding: A paradigm shift in computer vision from

recognition to reasoning". In: *AI Magazine* 45.3 (2024), pp. 429–435
https://doi.org/10.1002/aaai.12194 AI Magazine

Peer-Reviewed Workshop Papers and Extended Abstracts

[W8] Shivanand Kundargi, Ziwei Zhang, Kowshik Thopalli, and Tejas Gokhale. "On the Role of Mechanistic Interpretability for Vision-Language Prompt Learning". In: *Mechanistic Interpretability Workshop at ICML 2026*. 2026
https://icml.cc/virtual/2026/79343 ICML 2026 Workshop on Mechanistic Interpretability

[W7] Jessica RP Sutton, Thomas A Stanley, and Tejas Gokhale. "Using ground-based radar to investigate false detections in satellite precipitation". In: *106th AMS Annual Meeting*. AMS. 2026
https://ams.confex.com/ams/106ANNUAL/meetingapp.cgi/Paper/475997 AMS 2026

[W6] Yiran Luo, Joshua Feinglass, Tejas Gokhale, Kuan-Cheng Lee, Chitta Baral, and Yezhou Yang. "Grounding Stylistic Domain Generalization with Quantitative Domain Shift Measures and Synthetic Scene Images". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 7303–7313
https://arxiv.org/abs/2405.15961 CVPR 2024 Vision Datasets Understanding Workshop

[W5] Tejas Gokhale. "Towards robust visual understanding: from recognition to reasoning". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. 20. 2024, pp. 22665–22665
https://ojs.aaai.org/index.php/AAAI/article/view/30281 AAAI New Faculty Highlights

[W4] Tejas Gokhale, Joshua Feinglass, and Yezhou Yang. "Covariate shift detection via domain interpolation sensitivity". In: *First Workshop on Interpolation Regularizers and Beyond at NeurIPS 2022*. 2022
https://openreview.net/pdf?id=YkPjTHZDdm NeurIPS 2022 Interpolation Workshop

[W3] Kuldeep Kulkarni, Tejas Gokhale, Rajhans Singh, Pavan Turaga, and Aswin Sankaranarayanan. "Halluci-Net: Scene Completion by Exploiting Object Co-occurrence Relationships". In: *CVPR Workshop on AI for Content Creation*. 2021
https://arxiv.org/abs/2004.08614 CVPR 2021 AI for Content Creation Workshop

[W2] Tejas Gokhale, Shailaja Sampat, Zhiyuan Fang, Yezhou Yang, and Chitta Baral. "Cooking With Blocks: A Recipe for Visual Reasoning on Image-Pairs". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 2019, pp. 5–8
https://arxiv.org/abs/1905.12042 CVPR 2019 Vision Meets Cognition Workshop

[W1] Tejas Gokhale. "Vision beyond Pixels: Visual Reasoning via Blocksworld Abstractions". In: *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence,* 2019, pp. 6436–6437
https://www.ijcai.org/Proceedings/2019/0907.pdf IJCAI 2019 Doctoral Consortium

Technical Reports and Preprints

[P4] Sarah Rastegar, Violeta Chatalbasheva, Sieger Falkena, Anuj Singh, Yanbo Wang, Tejas Gokhale, Hamid Palangi, and Hadi Jamali-Rad. "InfSplign: Inference-Time Spatial Alignment of Text-to-Image Diffusion Models". In: *arXiv preprint arXiv:2512.17851* (2025)
https://arxiv.org/abs/2512.17851 Preprint

[P3] Ethan Wisdom, Tejas Gokhale, Chaowei Xiao, and Yezhou Yang. "Mole Recruitment: Poisoning of Image Classifiers via Selective Batch Sampling". In: *arXiv preprint arXiv:2303.17080* (2023)
https://arxiv.org/abs/2303.17080 Tech Report

[P2] Tejas Gokhale, Hamid Palangi, Besmira Nushi, Vibhav Vineet, Eric Horvitz, Ece Kamar, Chitta Baral, and Yezhou Yang. "Benchmarking spatial relationships in text-to-image generation". In: *arXiv preprint arXiv:2212.10015* (2022)
https://arxiv.org/abs/2212.10015 Tech Report

[P1] Maitreya Patel, Neeraj Varshney, Agneet Chatterjee, Tejas Gokhale, Yezhou Yang, Chitta Baral. Reliability-Checklist: Framework for Comprehensively Evaluating the Reliability of NLP Systems.
https://github.com/Maitreyapatel/reliability-checklist Tech Report

Books and Monographs

[B1] Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. *Advances in Multimodal Information Retrieval and Generation*. Synthesis Lectures on Computer Vision. Springer International Publishing, 2024. DOI: 10.1007/978-3-031-57816-8
https://link.springer.com/book/9783031578151 ISBN: 978-3-031-57815-1

Chapters:

- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. "Transformer-Driven Models for Language, Vision, and Multimodality". In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 11–34. ISBN: 978-3-031-57816-8. DOI: 10.1007/978-3-031-57816-8_2
- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. "Multimodal Information Retrieval". In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 35–91. ISBN: 978-3-031-57816-8. DOI: 10.1007/978-3-031-57816-8_3
- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. "Multimodal Content Generation". In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 93–134. ISBN: 978-3-031-57816-8. DOI: 10.1007/978-3-031-57816-8_4
- Man Luo, Tejas Gokhale, Neeraj Varshney, Yezhou Yang, and Chitta Baral. "Retrieval Augmented Modeling". In: *Advances in Multimodal Information Retrieval and Generation*. Cham: Springer International Publishing, 2025, pp. 135–157. ISBN: 978-3-031-57816-8. DOI: 10.1007/978-3-031-57816-8_5

Ph.D. Dissertation

[T1] Tejas Gokhale. 2023. Towards Reliable Semantic Vision. Order No. 30426752, Arizona State University.
https://www.proquest.com/docview/2813822780

Intellectual Property

[IP3] Tejas Gokhale, Sourajit Saha. *Zero-Shot Multimodal Retrieval with Multi-Scale Contextual Representations*. US Patent App. 64/045,450.

[IP2] Hamid Palangi, Besmira NUSHI, Vibhav Vineet, Eric J Horvitz, Semiha E KAMAR EDEN, and Tejas GOKHALE. *Automated evaluation of spatial relationships in images*. US Patent 12,444,106. 2025 [url]

[IP1] Tejas Gokhale, Rushil Anirudh, Jayaraman Thiagarajan, Bhavya Kailkhura, Chitta Baral, and Yezhou Yang. *Diversity using adversarially learned transformations for domain generalization*. US Patent App. 18/674,001. 2025 [url]

Funding

UMBC Cybersecurity Institute. ($50K)
Cybersecurity Graduate Fellows Program 2025

COEIT Innovations and Interdisciplinary Projects ($20K)
"Convergence Research: Resilient Next-Generation 911 Video Calling by Bridging 5G/6G Wireless Networks and Visual Intelligence" 2026-27

Sustainable Horizons Institute (via National Science Foundation Grant No. #2515632) ~$30K
Sustainable Research Pathways (SRP) Program 2026

DARPA SciFy (Scientific Feasibility) Program. ($3.8M)
"Modular Reasoning using Hybrid Inferential Formalisms" 2024-27

UMBC Center & Institute Departmentally-Engaged Research (CIDER). ($50K)
"Identification of Virga Precipitation Events" 2025-26

UMBC Cybersecurity Institute. ($45K)
Cybersecurity Graduate Fellows Program 2025

UMBC Strategic Awards for Research Transitions (START) ($25K)
"A Framework for Quantifying Typicality of AI-Generated Images" 2024-25

Maryland Procurement Office (via Johns Hopkins University) (~$30K)
"Modular Natural Language Understanding" (PI: Frank Ferraro) 2024

UMBC Summer Research Faculty Fellowship (SURFF) ($8K)
"Improving the Continual Learning Ability of Visual Recognition Systems via Targeted Unlearning" 2024

Microsoft Research Accelerate Foundation Models Academic Research ($20K)
Cloud Computing and OpenAI Credits 2024

Google Cloud (~$2.5K)
Education Credits 2023-24

Teaching

Instructor, UMBC

CMSC 472/672 (prev. 491/691) Computer Vision [website] Fall 2026, Fall 2025, Spring 2024, Fall 2023
CMSC 475/675 Neural Networks [website] Spring 2026, Spring 2025
CMSC 491/691 Robust Machine Learning [website] Fall 2024
CMSC 898 Pre-Doctoral Candidacy Research
CMSC 799 Master's Thesis Research
CMSC 699 Independent Study
CMSC 499 Independent Study

Graduate Teaching Associate, Arizona State University

CSE310: Data Structures & Algorithms Spring 2020
CSE408: Multimedia Information Systems Spring 2019
CSE110: Introduction to Programming, Fall 2018

Guest Lecturer, Arizona State University

CSE598, Perception in Robotics — Spring 2022
CSE408, Multimedia Information Systems — Spring 2019

Student Instructor, BITS Pilani Goa Campus

CTE: Advanced Image Processing — Spring 2015

Students

PhD

- Sourajit Saha (M.S. UMBC) — Ph.D. CS [current], UMBC
  - *Internships: SCALE 2026 & SCALE 2024 (at JHU HLTCOE)*
  - *ACL Great Reviewer Award, 2026*
  - *CVPR Outstanding Reviewer Award, 2026*
  - *UMBC 3 Minute Thesis Finalist, 2026*
  - *WACV 2025 Broadening Participation Scholarship*
  - *GSA Professional Development Grant, 2025*
  - *Lambda Research Grant Program 2025*
- Ziwei Zhang (M.S. USTC, China) — Ph.D. CS [current], UMBC
  - *CSEE Summer Research Fellowship, 2025*
- Shivanand Kundargi (B.S. KLE U., India) — Ph.D. CS [current], UMBC
  - *Internship: Lawrence Livermore National Laboratory, 2025*
  - *UMBC Cyber Graduate Fellow, 2025*
  - *GSA Professional Development Grant, 2026*
  - *CoLLAs Conference Travel Award, 2026*
  - *CSEE Summer Research Fellowship, 2025 (declined)*
- Naren Sivakumar (M.S. UMBC) — Ph.D. CS [current], UMBC
  - *Internship: SCALE 2026 (at JHU HLTCOE)*
  - *CSEE Summer Research Fellowship, 2026*
- Dylan Lang (M.S. ASU) — Ph.D. CS [current], UMBC

PhD (as Committee Member)

- Yuxuan Jiang (advisor: Frank Ferraro) — Ph.D. CS [current], UMBC
- Jacob Rubinstein (advisor: Don Engel) — Ph.D. CS [current], UMBC
- Nilay Yilmaz (advisor: Yezhou Yang) — Ph.D. CS [current], ASU
- Shubhasis Roy Dipta (advisor: Frank Ferraro) — Ph.D. CS [current], UMBC
- Agneet Chatterjee (advisor: Yezhou Yang and Chitta Baral) — Ph.D. CS [current], ASU
- Mark Jarzynski (advisor: Marc Olano) — Ph.D. CS [current], UMBC
- Yiran Luo (advisor: Chitta Baral and Yezhou Yang) — Ph.D. CS [current], ASU
- Maitreya Patel (advisor: Yezhou Yang and Chitta Baral) — Ph.D. CS 2026, ASU
- Sheng Cheng (advisor: Yezhou Yang) — Ph.D. CS 2025, ASU [dissertation]

MS Thesis

– Nathaniel Therrien — M.S. CS [current], UMBC

MS Thesis (as Committee Member)

– Avi Donaty (advisor: Don Engel) — M.S. CS 2026, UMBC
– Beamlak Bekele (advisor: Don Engel) — M.S. CS 2026, UMBC
– Joey Mule (advisor: Riadul Islam) — M.S. CE 2026, UMBC
– Naomi Angela Tack (advisor: Don Engel) — M.S. CS 2024, UMBC

Other MS/PhD

– Research Assistant: Joey Mule — Summer 2026
– Independent Study: Neel Patel — Spring 2025, Fall 2024
– Independent Study: Shaswati Saha — Ph.D. CS [current], UMBC
– Independent Study: Varun Magotra — Spring 2024

Undergraduate

– Summer Visitor: Arunabh Ramesh (Rensselaer Polytechnic Institute) — Summer 2026
– Independent Study: Nicholas Harrell — Spring 2025
– Independent Study: Alexander Shaner — Spring 2025
– Visitors: Alexander Haibel — 2025-2026
– Visitors: Tetevi Wilson, Dhanush Bharadwaj — 2024-2025
– Visitors: Joey Mule, Luke Parrish — 2023-2024
– UMBC CWIT Scholar: Chloe Wood — 2024-25
– UMBC CWIT Scholar: Danielle Burton — 2023-24

Teaching Assistants

– Aydin Ayanzadeh, CMSC 475/675 Neural Networks — Spring 2026
– Yu Liu, CMSC 472/672 Computer Vision — Fall 2025
– Ziwei Zhang, CMSC 475/675 Neural Networks — Spring 2025
– Sourajit Saha, CMSC 491/691 Computer Vision — Spring 2024
– Aidin Shiri, CMSC 491/691 Computer Vision — Fall 2023

Ph.D. Mentees (at ASU)

– Maitreya Patel — Ph.D. CS [current], ASU
– Agneet Chatterjee — Ph.D. CS [current], ASU
– Nilay Yilmaz — Ph.D. CS [current], ASU

MS (Thesis) Mentees (at ASU)

– Maitreya Patel — M.S. CS 2022, ASU [thesis]
– Abhishek Chaudhary — M.S. CS 2021, ASU [thesis]

Undergraduate Mentees (at ASU)

– ASU FURI Program: Mertay Dayanc — B.S CS, 2020
– CS Capstone: Paul Butler, Jace Lord, Aashwin Ranjan, Sagarika Pannase, William Tith — 2019-20

Academic Service

National Science Foundation

– Panel, IIS/III 2026, 2025
– Panel, GEO 2026
– Reviewer, GRFP 2025

Tutorial Chair, International Conference on Computer Vision (ICCV) 2025

Area Chair / Action Editor

– International Conference on Computer Vision (ICCV) 2025
– Advances in Neural Information Processing Systems (NeurIPS) 2024-25
– NeurIPS Position Papers Track 2025
– Winter Applications of Computer Vision (WACV) 2025-2027
– Association for Computational Linguistics (ACL) 2024
– North American Chapter of the ACL (NAACL) 2024
– Empirical Methods in Natural Language Processing (EMNLP) 2024

Session Chair

– Winter Applications of Computer Vision (WACV) 2025

Reviewer / Program Committee:

– Conference on Computer Vision and Pattern Recognition (CVPR) 2023-25
– International Conference on Computer Vision (ICCV) 2023
– European Conference on Computer Vision (ECCV) 2022–26
– Winter Conference on Applications of Computer Vision (WACV) 2021–24
– International Conference on Machine Learning (ICML) 2023–25
– Advances in Neural Information Processing Systems (NeurIPS) 2022–24
– International Conference on Learning Representations (ICLR) 2022–25
– AAAI Conference on Artificial Intelligence (AAAI) 2021–24
– AAAI Senior Member Presentation Track 2026
– Conference on Language Models (COLM) 2024
– Association for Computational Linguistics (ACL) 2021–26
– Empirical Methods in Natural Language Processing (EMNLP) 2021–23
– North American Chapter of the ACL (NAACL) 2021–23
– International Conference on Robotics and Automation (ICRA) 2019–2023
– International Conference on Intelligent Robots and Systems (IROS) 2022
– IEEE Robotics and Automation Letter (RA-L) 2020-24
– IEEE Transactions of Pattern Analysis and Machine Intelligence (T-PAMI) 2024–25
– ACM Multimedia 2025
– ACM Transactions of Computing for Healthcare 2024
– ACM Computing Surveys 2024
– Springer Machine Vision and Applications (MVAP) 2020
– Springer Book Proposals Reviewer 2024
– Mentor, AAAI Undergraduate Student Consortium 2024
– Award Committee, Best Student Abstract, AAAI 2024

Leadership:

– Organizer, Tutorial on Responsibly Building Generative Models [Website] ECCV'24
– Team Lead, Summer Camp for Applied Language Exploration (SCALE) 2024, JHU Human Language Technology Center of Excellence [Website] Summer 2024

– Organizer, Tutorial on Reliability of Generative Models in Vision [Website] WACV'24
– Organizer, Workshop on Open-Domain Reasoning under Multi-Modal Settings (ODRUM), [Website] [YouTube] CVPR'23
– Organizer, Workshop on Open-Domain Retrieval under Multi-Modal Settings (ODRUM), [Website] [YouTube] CVPR'22
– Organizer, Tutorial on Semantic Data Engineering under Multimodal Settings (SERUM) [Website] WACV'23
– Organizer, 2021 Frontiers of V&L Seminar Series, [Website], [YouTube] ASU

Conference Volunteer

– Volunteer, 2019 Southwest Robotics Symposium, Tempe AZ SWRS 2019
– Volunteer, International Conference on Machine Learning 2020

Misc:

– Lead Vocalist, CVPR House Band 2024, 2025

University Service

University Service (at UMBC):

University-Level

– UMBC Strategic Planning Subcommittee on Research Computing and Infrastructure 2026–
– UMBC HPCF Governance, Subcommittee for CHIP-GPU 2025–
– Search Committee, Faculty Development Center Director 2026
– Reviewer, UMBC ORCA Internal Grants 2025
– Judge, UMBC Annual Research Symposium 2026
– PPR Seminar: Advances in Perception, Prediction, and Reasoning [Website]
– Faculty Learning Community, UMBC Faculty Development Center 2025–26
– Faculty Learning Community, UMBC Faculty Development Center 2024–25
– Member, Asian and Asian American Faculty Staff Council 2023–

College-Level

– Panelist, Exploring Your Options: What Does Research Look Like in COEIT? Summer 2026
– Reviewer, COEIT Cybersecurity Research and Education Proposals Fall 2024
– Interviewer, COEIT Staff Searches 2025
– Reviewer, COEIT Student Summer Project Award Summer 2025
– Faculty Mentor, Center for Women in Technology 2023–2025
– Faculty Volunteer, COEIT Ph.D. Open House 2026, 2025, 2024
– COEIT Admitted Student Call Campaign 2026

Department-Level

– Course Development, CMSC 475/675: Neural Networks [Website]
– Course Development, CMSC 491/691: Robust Machine Learning [Website]
– Course Development, CMSC 472/672: Computer Vision [Website]
– Regular Graduate Course Proposal, CMSC 672: Computer Vision Approved
– Regular Undergraduate Course Proposal, CMSC 472: Computer Vision Approved
– CSEE Computer Science Undergraduate Committee 2025–

- CSEE Computer Science Graduate Admissions Committee 2023–25
- CSEE Publicity Committee 2023–25
- CSEE Computer Science Undergraduate Student Advisor 2023–
- Faculty Mentor, UMBC IEEE + Tau Beta Pi Open Lab Fall 2024
- CSEE Lightning Talks and Open House Fall 2026, Fall 2025, Fall 2024, Fall 2023
- Reviewer, CSEE Research Day Spring 2026, Spring 2024
- Interviewer, CSEE Faculty Candidates 2024, 2025

University Service (at ASU):

- Founder, Summer Vision Reading Group, ASU [Website]
- Course Development, CSE591: Frontier Topics in Vision & Language [YouTube] [website] Sp.2021, ASU
- Founding Advisor, ASU Machine Learning Club, ASU
- Award Reviewer, GPSA Teaching Award Reviewer ASU
- Mentor, Graduate Student Mentorship Program, ASU
- Project Mentor, CSE598 - Perception in Robotics, ASU Spring 2022
- Project Mentor, CSE576 - Natural Language Processing, ASU Fall 2018

Presentations

(Invited Talk) CSEE Research Day, UMBC 04/2026
"Cognitive Vision: Intelligence as Concept Learning"

(Invited Talk) Amazon Science 03/2026
"Concept-Based Understanding: A Tale of Four 'R's "

(Invited Talk) Lawrence Livermore National Laboratory, CASC Seminar 03/2026
"Cognitive Vision: Intelligence as Concept Learning"

(Keynote) Winter Applications of Computer Vision, Event Vision Workshop 03/2026
"Concept-Based Event Understanding: A Multi-Modal Perspective"

(Invited Talk) International Conference on Computer Vision, AC Workshop 10/2025
"Active Data Pursuit for Robust Vision"

(Invited Talk), UMBC COEIT Research Day 04/2025
"Cognitive Vision: Concepts, Contexts, and Semantics"

(Invited Talk), UMBC Information Systems Seminar 11/2024
"Cognitive Vision: Concepts, Contexts, and Semantics"

(Tutorial), European Conference on Computer Vision 10/2024
"Evaluation and Benchmarking for Text-to-Image Models"

(Lightning Talk), IARPA Video-LINCS Proposers Day 02/2024
"Robust Visual Understanding: Knowledge-Guided and Multimodal Reasoning"

(Tutorial), Winter Conference on Applications of Computer Vision 01/2024
"Challenges with Evaluation of Text-to-Image Models" [website]

(Invited Talk), PRG Seminar, UMIACS (University of Maryland) 11/2023
"Robust Visual Understanding in the Multimodal Era"

(Invited Talk) *"Towards Reliable Semantic Vision"* Spring 2023

– Rochester Institute of Technology (02/23)
– Binghamton University (03/23)
– University of Maryland Baltimore County (03/23)
– Indiana University (03/23)
– Case Western Reserve University (03/23)
– Colorado School of Mines (03/23)
– Temple University (04/2023)

(Tutorial), Winter Conference on Applications of Computer Vision — 01/2023
"Semantic Data Engineering for Robustness Under Multimodal Settings" [website]

(Invited Talk) University of Illinois at Chicago — 10/2022
"Robust Semantic Vision"

(Invited Talk) Microsoft Research — 10/2022
"Benchmarking Spatial Relationships in Text-to-Image Generation"

(Doctoral Consortium) CVPR, New Orleans — 06/2022
"Discovering Transformations for Generalization in Semantic Vision"

(Guest Lecture) Arizona State University CSE 598 — 03/2022
"Introduction to Generalization in Semantic Vision"

(Invited Talk) Arizona State University ML Club — 09/2021
"Robust Visual Understanding"

(Doctoral Consortium), IJCAI, Macao — 08/2019
"Vision Beyond Pixels"

(Tutorial) Telluride Neuromorphic Cognition Engineering Workshop, — 07/2019
"Reasoning about Objects and Actions via Block-Play" [website]

(Invited) Birla Institute of Technology and Science (BITS Pilani) — 04/2018
"Deep Learning Methods in Imaging and Computer Vision"

Awards

- AAAFSC Inaugural Community Builder Award, UMBC — 2026
- UMBC Career Connector Award — 2026
- CVPR 2024 VDU Workshop, Best Paper Award — 2024
- Research Excellence Award, ASU GPSA — 2022
- Outstanding Mentor Award, ASU GPSA — 2022
- NeurIPS Top Reviewer — NeurIPS 2022
- CVPR 2022 Doctoral Consortium — CVPR 2022
- ICLR Best Reviewer — ICLR 2022
- SCAI Doctoral Fellowship (ASU), — 2022, 2021, 2020
- Engineering Graduate Fellowship, (ASU Engineering) — 2023, 2020
- ASU GPSA Travel Award — for WACV 2023
- Graduate College Travel Award (declined) — WACV'23, CVPR'22
- Graduate College Travel Award (accepted) — ICCV'21, EMNLP'20, ECCV'20
- IJCAI 2019 Doctoral Consortium — IJCAI 2019
- Inducted, IEEE Eta Kappa Nu, Sigma Chapter — CMU, 2017

- National Talent Scholarship, National Council of Educational Research and Training (Govt. of India) 2007–2015

Media

- DARPA's AI is built to call BS on wild weapons claims
  Scientific American 04/2026
- UMBC team leads research into AI tools that can assess the feasibility of scientific claims
  UMBC News 04/2025
- Alum inspires next generation of computer vision researchers
  ASU Full Circle 10/2024
- Frontiers of multimodal learning: A responsible AI approach
  Microsoft Research Blog 09/2023
- CASC research in ML robustness debuts at AAAI conference
  News and Press, LLNL Computing 02/2021
- HuggingFace and Intel release a solution for high-fidelity text and image consistency
  NetEase (163.com), China 04/2024

References

Available upon request